

COURSE TITLE : MICROCONTROLLER AND INTERFACING LAB
COURSE CODE : 4217
COURSE CATEGORY : A
PERIODS/WEEK : 6
PERIODS/SEMESTER: 84
CREDITS : 3

On completion of this course the student will be able to

1. Familiarization of Microcontroller Kit
2. Multiplication of two 8 bit numbers
3. Division of two numbers
4. Finding the largest data in an array
5. Find number of occurrence of a data in an array
6. Sorting: ascending and descending
7. BCD to Hex conversion
8. Hex to BCD conversion
9. Binary to ASCII
10. ASCII to Binary
11. Square wave generation using port
12. Counter using timer
13. Program using interrupt
14. Least Common Multiple
15. Greatest Common Divisor
16. Sum of first N natural numbers
17. Celsius scale to Fahrenheit scale
18. Sum of a given array
19. Average of an array
20. String comparison
21. Pressure unit conversions
22. To generate an arithmetic expression
23. Generation of sequence

Interfacing with application boards

24. Digital I/O
25. Matrix keyboard
26. Seven segment displays
27. LCD Displays
28. Traffic light
29. 8 bit ADC
30. 8 bit DAC
31. Stepper motor control
32. DC motor control
33. Sending data through serial port of controller
34. Temperature IC interfacing